

LINE-BY-LINE CODE MASTER REFERENCE

Exhaustive Line-by-Line Breakdown: my_practice_v2.py

Project: Universal Customer Churn & Retention Prediction System

Section 1: Imports

```
import pandas as pd
```

Loads Pandas library with standard alias pd for 2D tabular DataFrame manipulation.

```
from sklearn.preprocessing import StandardScaler
```

Imports StandardScaler to perform Standardization ($z = \frac{x - \mu}{\sigma}$), scaling numbers to Mean = 0.

```
from sklearn.model_selection import train_test_split
```

Splits data into 80% Training Set and 20% Testing Set.

```
from sklearn.linear_model import LogisticRegression
```

Imports binary classification model algorithm using Sigmoid S-Curve activation.

```
from sklearn.metrics import accuracy_score, confusion_matrix,  
classification_report
```

Imports metrics tools to compute Accuracy, Confusion Matrix (TP, TN, FP, FN), and Precision/Recall.

Section 2: Data Cleaning & Imputation

```
df = pd.read_csv('data/customer_churn_data.csv')
```

Loads 1,000 customers x 11 columns dataset into DataFrame df.

```
median = df['Total_Charges_INR'].median()
```

Calculates outlier-proof 50th percentile median value for monetary column.

```
df['Total_Charges_INR'] =  
df['Total_Charges_INR'].fillna(median)
```

Fills all 20 NaN missing cells with calculated median value.

Section 3: Categorical Encoding & Scaling

```
df_encoded = pd.get_dummies(df, columns=cat_cols,  
drop_first=True)
```

Converts text strings into 0/1 binary columns. drop_first=True prevents Dummy Variable Trap.

```
df_encoded[num_cols] =  
scale.fit_transform(df_encoded[num_cols])
```

Calculates Mean/Std Dev and rescales numeric features so Mean = 0.

Section 4: Train-Test Split & Fitting

```
X_train, X_test, y_train, y_test = train_test_split(x, y,  
test_size=0.2, random_state=41)
```

Splits 800 training rows and 200 testing rows. Maintains strict positional unpacking sequence.

```
test.fit(X_train, y_train)
```

Model runs Gradient Descent on 800 training customers to optimize feature weights (w) .

Section 5: Advanced Tasks (Probabilities & Threshold)

```
y_proba = test.predict_proba(X_test)
```

Computes exact decimal probabilities $(P(\text{Stay}), P(\text{Churn}))$ for every test customer.

```
y_pred_custom = (test.predict_proba(X_test)[:, 1] ≥  
0.30).astype(int)
```

Applies custom business threshold of 0.30 (30% risk alert), boosting Recall from 63% to 83.56%!